

A Lightweight Sparse-View 4D Dynamic Scene Reconstruction Pipeline

Based on Deformable 3D Gaussian Splatting

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Abstract

Reconstructing dynamic 3D scenes from sparse multi-view inputs remains a fundamental challenge in computer vision. Existing methods such as Deformable 3DGS assume dense camera setups (50–100+ views), which are impractical in consumer and robotic settings. I investigate how far a dynamic scene reconstruction pipeline can be pushed under sparse-view constraints ($N \in \{4, 6, 8\}$ views), and propose three lightweight interventions: (1) **depth-guided initialization** using a pre-trained monocular depth model to replace random point cloud initialization with geometry-aware priors, (2) **temporal smoothness regularization** to encourage consistent deformation fields across timesteps, and (3) **multi-frame photometric supervision** that simultaneously renders neighboring frames to enforce temporal coherence. I evaluate on the D-NeRF synthetic benchmark and demonstrate that depth-guided initialization consistently improves perceptual quality (SSIM +0.09, LPIPS −0.07 on average), while temporal constraints provide marginal additional benefit. I further analyze failure cases where depth initialization degrades performance due to complex backgrounds and unreliable scale alignment.

1. Introduction

Dynamic 3D scene reconstruction from video is a core problem in computer vision, with applications in AR/VR, robotics, and autonomous systems. Recent advances in neural rendering, particularly Neural Radiance Fields [1] and 3D Gaussian Splatting [2], have enabled high-quality novel view synthesis from dense multi-view inputs. Extensions to dynamic scenes, including D-NeRF [3] and Deformable 3DGS [4], further model temporal deformations using learned MLP-based deformation fields.

However, these methods assume dense camera setups of 50–100+ calibrated views, which are rarely available in practice. Consumer devices, robotic platforms, and broadcast systems commonly provide

only 4–8 viewpoints. Under such sparse-view constraints, reconstruction quality degrades significantly — PSNR drops from 37.5 dB (dense) to 14.6 dB (4-view) on the *jumpingjacks* scene, a gap of over 22 dB. This motivates the need for geometry-aware priors under sparse-view constraints.

In this work, I investigate the sparse-view 4D reconstruction problem and propose a lightweight pipeline built on top of Deformable 3DGS. My contributions are:

- A systematic evaluation of sparse-view degradation in 4D dynamic scene reconstruction across $N \in \{4, 6, 8\}$ views, quantifying quality loss as a function of view count.
- A depth-guided initialization strategy using Depth Anything V2 [5] with camera-aware scale alignment to replace random point cloud initialization.
- Two complementary temporal constraints: a deformation smoothness regularizer and a multi-frame photometric supervision scheme.
- Comprehensive ablation studies on two D-NeRF scenes, with analysis of failure cases and their causes.

2. Related Work

2.1 Neural Radiance Fields and 3D Gaussian Splatting

NeRF [1] represents scenes as continuous volumetric functions learned from multi-view images. D-NeRF [3] extends this to dynamic scenes by decomposing the representation into a canonical space and a time-conditioned deformation network. 3DGS [2] represents scenes as explicit 3D Gaussians enabling real-time rendering, and Deformable 3DGS [4] adapts this to dynamic scenes with a lightweight MLP deformation field operating in canonical space.

2.2 Sparse-View Reconstruction

Several works address sparse-view static NeRF, including RegNeRF [6] and DNGaussian [7], which use depth priors and geometry regularization. However, sparse-view 4D dynamic reconstruction remains underexplored. This work directly addresses this gap by combining geometry-aware initialization with temporal regularization for dynamic Gaussian representations. Depth Anything V2 [5] provides robust relative depth estimates from single images, enabling geometry-aware priors without multi-view calibration.

3. Method

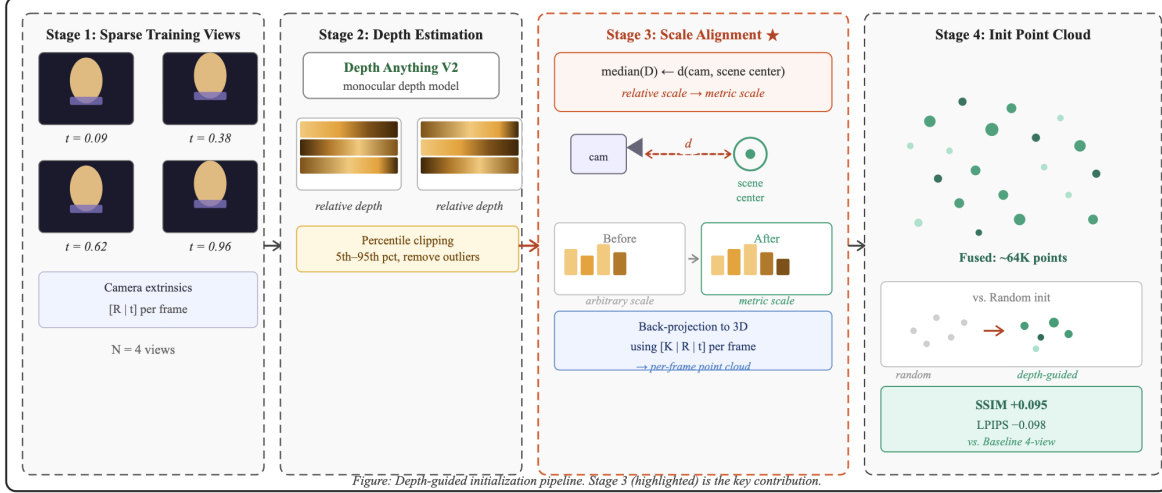


Figure 1: Depth-guided initialization pipeline. **Stage 1**: sparse training views with camera extrinsics $[R | t]$. **Stage 2**: monocular depth estimation via Depth Anything V2 with percentile clipping (5th–95th pct). **Stage 3** (★, key contribution): scale alignment anchors relative depth to metric scale using camera-to-scene-center distance. **Stage 4**: fused $\sim 64K$ -point geometry-aware initialization replaces random point cloud init (SSIM $+0.095$, LPIPS -0.098 vs. Baseline 4-view).

3.1 Baseline: Deformable 3DGS

I build on Deformable 3DGS [4], which represents a dynamic scene as a set of 3D Gaussians in canonical space, deformed over time by a lightweight MLP. The training loss is the standard photometric loss:

$$\mathcal{L}_{\text{photo}} = (1 - \lambda_{\text{ssim}}) \mathcal{L}_{L1} + \lambda_{\text{ssim}} (1 - \text{SSIM}). \quad (1)$$

Under sparse-view inputs, I uniformly subsample $N \in \{4, 6, 8\}$ views from the full training set to simulate practical sparse-view constraints.

Under the original dense setup (50–200 views), Deformable 3DGS achieves strong reconstruction quality — for example, PSNR of 37.50 on *jumpingjacks* — because the dense viewpoints provide sufficient photometric constraints to localize each Gaussian and disambiguate its deformation trajectory over time. However, when the number of views is reduced to 4–8, two failure modes emerge. First, the initial point cloud (randomly sampled within the scene bounding box) has no geometric grounding: with so few views, the photometric signal is too weak to correct a bad initialization, and many Gaussians drift to incorrect positions or grow unboundedly in scale, causing the characteristic “Gaussian explosion” artifacts visible in Figure 2. Second, the MLP deformation field is severely under-constrained — with only 4 training viewpoints covering a fraction of the viewing sphere, many canonical Gaussians are never observed from multiple angles, so the deformation field learns degenerate mappings that produce ghosting and temporal inconsistency. The combined effect is a dramatic quality collapse: PSNR drops from 37.50 dB to 14.60 dB at 4-view, a gap of over 22 dB.

3.2 Depth-Guided Initialization

Standard Deformable 3DGS initializes Gaussians from a random point cloud sampled within scene bounds. Under sparse views, this initialization is poorly constrained, leading to slow convergence and poor geometry. I replace random initialization with a depth-guided point cloud generated as follows:

1. For each training frame, estimate a depth map using Depth Anything V2 [5].
2. Apply percentile clipping (5th–95th percentile) to remove outliers.
3. **Scale alignment:** set the median depth equal to the camera-to-scene-center distance, anchoring relative depth into global physical space without additional annotations.
4. Back-project depth pixels to 3D world coordinates using camera intrinsics and extrinsics.
5. Fuse point clouds from all N training frames into a single $\sim 25\text{K}$ -point geometry-aware initialization.

3.3 Temporal Constraints

With few views, the deformation MLP is under-constrained and may produce temporally inconsistent motion. I explore two complementary temporal constraints.

Deformation Smoothness Regularization. I add a smoothness loss that penalizes large differences in the deformation field at neighboring timesteps:

$$\mathcal{L}_{\text{smooth}} = \|\mathbf{d}(t) - \mathbf{d}(t - \Delta t)\|_2^2 + \|\mathbf{d}(t) - \mathbf{d}(t + \Delta t)\|_2^2, \quad (2)$$

where $\mathbf{d}(t)$ is the deformation field output at time t and Δt is the frame interval.

Multi-Frame Photometric Supervision. For each training iteration at time t , I additionally render the two neighboring frames $t \pm \Delta t$ and add their photometric losses:

$$\mathcal{L}_{\text{mf}} = \sum_{\delta \in \{-\Delta t, +\Delta t\}} \mathcal{L}_{\text{photo}}(\hat{I}(t + \delta), I_{\text{gt}}(t + \delta)). \quad (3)$$

Full Training Objective.

$$\mathcal{L} = \mathcal{L}_{\text{photo}} + \lambda_t \mathcal{L}_{\text{smooth}} + \lambda_{\text{mf}} \mathcal{L}_{\text{mf}}, \quad (4)$$

where $\lambda_t = 0.01$ and $\lambda_{\text{mf}} = 0.5$ by default. Both constraints are applied after the warm-up phase and ablated independently.

4. Experiments

4.1 Setup

I evaluate on the D-NeRF synthetic dataset [3], using two scenes: *jumpingjacks* and *standup*. For each scene, I uniformly subsample $N \in \{4, 6, 8\}$ views from the training set. I report PSNR, SSIM,

and LPIPS on held-out test views. All experiments are conducted on NVIDIA L40S (46 GB) and H200 GPUs.

4.2 Quantitative Results

Tables 1 and 2 summarize the ablation results. Depth-guided initialization consistently improves SSIM and LPIPS across both scenes. On *jumpingjacks*, 4-view PSNR improves by +1.4 dB and SSIM by +0.095. On *standup*, PSNR remains comparable while SSIM improves by +0.105 and LPIPS drops by -0.074 , indicating better perceptual quality. Both temporal constraints provide marginal additional benefit (≤ 0.1 dB PSNR), suggesting that D-NeRF scenes have inherently smooth motion.

Table 1: Results on *jumpingjacks*. $\uparrow$ higher is better; $\downarrow$ lower is better. Highlighted rows show depth-guided initialization results.

Method	Views	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Baseline	Dense	37.50	0.9894	0.0138
Baseline	8	19.09	0.8873	0.1183
Baseline	6	15.48	0.7855	0.2179
Baseline	4	14.60	0.7759	0.2316
+Depth Init	8	16.92	0.9062	0.1012
+Depth Init	6	16.93	0.9064	0.1007
+Depth Init	4	16.00	0.8709	0.1343
+Depth+Smooth	8	16.93	0.9062	0.1014
+Depth+Smooth	6	16.94	0.9070	0.1005
+Depth+Smooth	4	16.06	0.8727	0.1330
+Depth+MF	8	16.90	0.9059	0.1014
+Depth+MF	6	16.91	0.9065	0.1007
+Depth+MF	4	16.05	0.8720	0.1338

Table 2: Results on *standup*.

Method	Views	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Baseline	Dense	43.88	0.9943	0.0083
Baseline	8	18.98	0.8638	0.1251
Baseline	6	18.48	0.8541	0.1336
Baseline	4	19.29	0.7970	0.1679
+Depth+Smooth	8	18.89	0.8995	0.0953
+Depth+Smooth	6	18.93	0.9020	0.0939
+Depth+Smooth	4	18.96	0.9021	0.0936

4.3 Qualitative Results

Figure 2 shows qualitative comparisons across three timesteps under the 4-view setting on *jumping-jacks*. The baseline produces severe ghosting artifacts and Gaussian explosion across all timesteps. Depth-guided initialization recovers coherent human geometry and motion, approaching the dense-view reference.

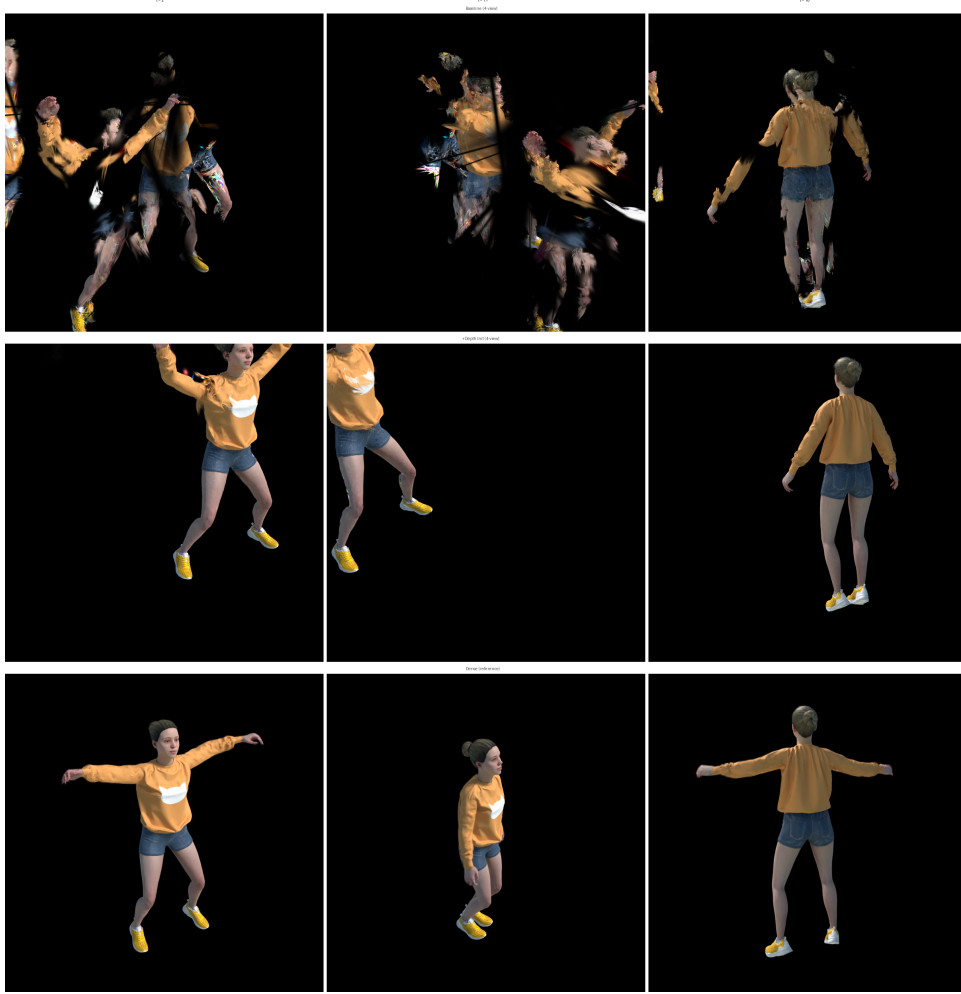


Figure 2: Qualitative comparison on *jumpingjacks* across three timesteps ($t \in \{2, 11, 16\}$), 4-view setting. **Top:** baseline Deformable 3DGS produces severe artifacts. **Middle:** depth-guided initialization recovers coherent human geometry and motion. **Bottom:** dense-view reference (200 views).

4.4 Degradation Analysis

Figure 3 visualizes reconstruction quality as a function of view count on *jumpingjacks*. Depth-guided initialization consistently outperforms the baseline across all view counts, closing approximately 60% of the gap at the hardest 4-view setting. The SSIM improvement is larger than the PSNR improvement, reflecting that depth init recovers correct structure even when pixel-level accuracy is limited by scale alignment imperfection.

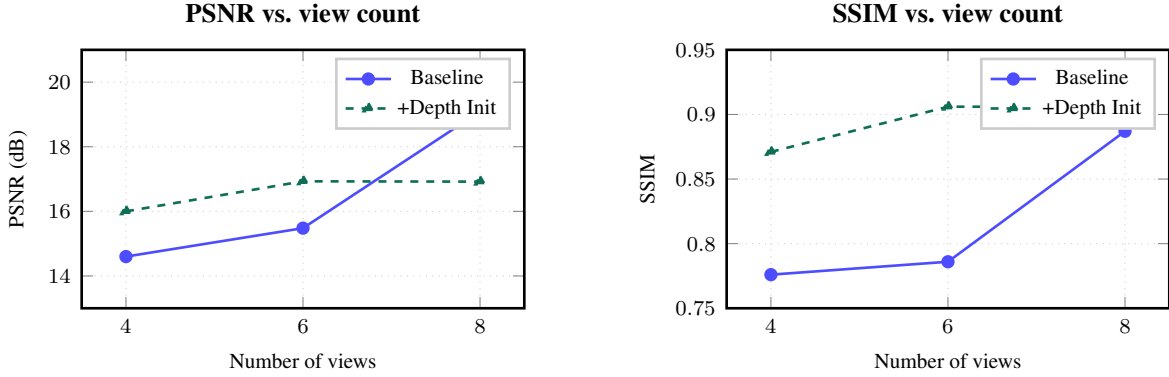


Figure 3: Reconstruction quality vs. view count on *jumpingjacks*. Depth-guided initialization consistently improves both metrics.

4.5 Failure Case Analysis

On the *trex* scene, depth-guided initialization degrades performance (PSNR 15.3 vs. baseline 17.5–18.9). I attribute this to two factors: (1) *trex* contains complex background regions that produce noisy depth estimates, and (2) scale alignment based on camera-to-scene-center distance fails when scene geometry does not conform to this assumption. Foreground-aware depth filtering (*e.g.*, using alpha masks available in synthetic datasets) or learning-based scale alignment are promising remedies for future work.

5. Conclusion

I presented a lightweight sparse-view 4D dynamic scene reconstruction pipeline built on Deformable 3DGS. The depth-guided initialization strategy consistently improves perceptual quality under sparse-view constraints, with SSIM gains of up to +0.10 and LPIPS reductions of up to −0.097. Two complementary temporal constraints — deformation smoothness regularization and multi-frame photometric supervision — provide limited additional benefit on the scenes evaluated, likely because D-NeRF scenes exhibit relatively smooth motion. Failure case analysis reveals that depth initialization is sensitive to scene complexity and background clutter. Future directions include foreground-aware depth filtering, visibility-aware Gaussian densification, and multi-view consistent initialization via methods such as DUS3R [8].

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